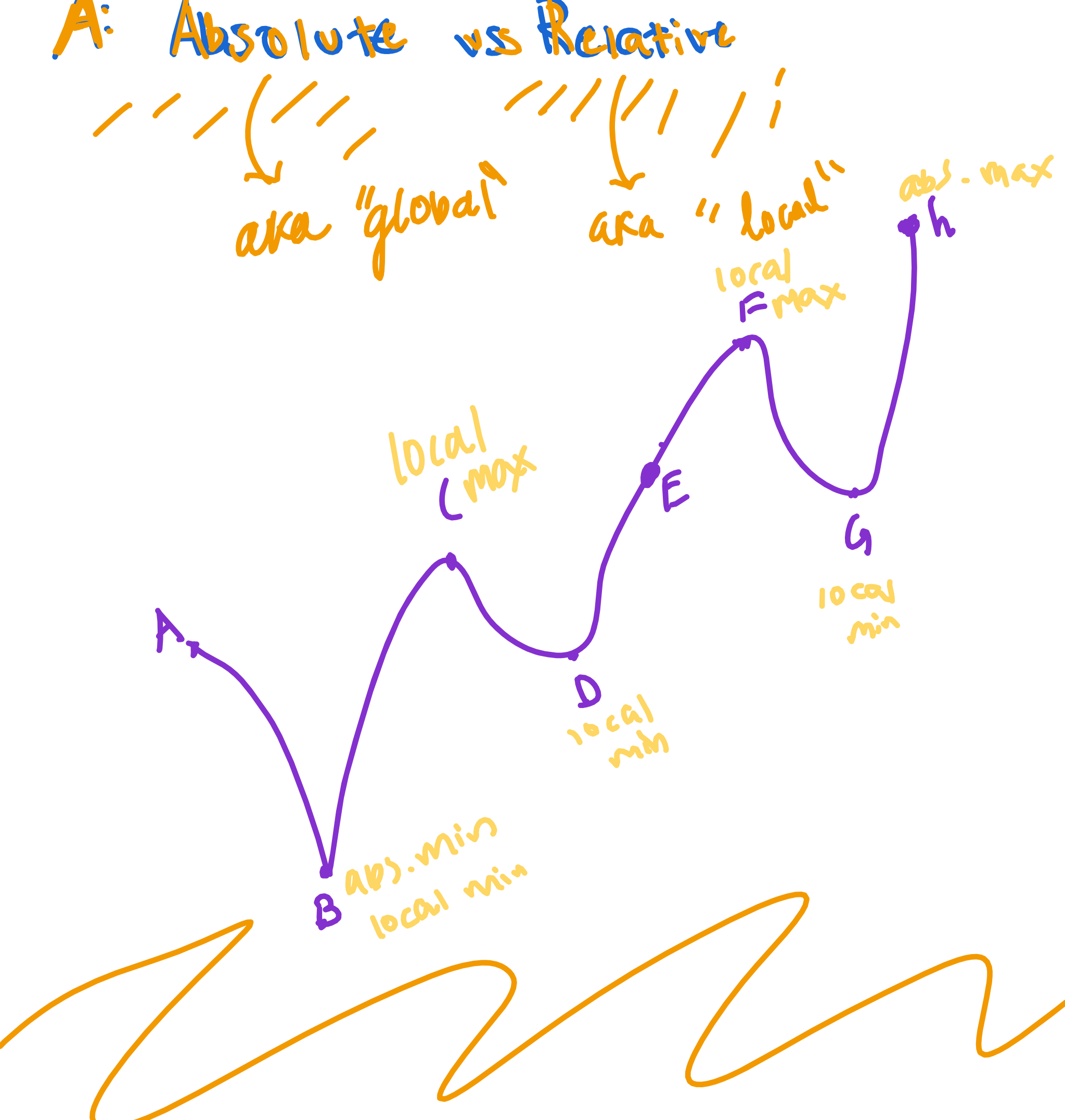


Extrema on an Interval

Section 3.1

9/22/25

A: Absolute vs Relative



B: Critical Points: (potential extrema)

1) $f'(x) = 0$

or

2) $f'(x) = \text{DNE}$

Candidate's test: absolute extrema compare y-values of critical points & end points

$f(x) = 3x^2 - 24x - 1$ on $[-1, 5]$

$f'(x) = 6x - 24 = 0$

$x = 4$ is on $[-1, 5]$ ✓

absolute max is at $(1, 26)$

abs min: $(4, -49)$

x	f(x)
-1	26
4	-49
5	-46

Ex 2 $f(x) = 3x^{2/3} - 2x + 1$ on $[-1, 8]$

$f'(x) = 2x^{-1/3} - 2 = 0$

$\frac{2 - 2\sqrt[3]{x}}{\sqrt[3]{x}} = 0$

$x = 0$ $x = 1$

x	f(x)
-1	6
0	1
1	2
8	-3

abs max: $(-1, 6)$
abs min: $(8, -3)$